

ANSWERS

PAGE 26–27: BIG NUMBER QUIZ

1. Two. They're the ones you took!
2. Numbers for a front door.
3. Read the first sentence in the question again for the answer.
4. 100. Think about it...
5. Just over an hour.
6. Just the 4 dead crows. The rest flew away when they heard the gunshots.
7. 12 kg
8. Because 1 hour 20 minutes is the same as 80 minutes.
9. None – it was Noah who built the Ark, not Moses!
10. Three
11. One – the first one!
12. 63
13. There is no missing £1 – it's a trick question. The question says "each customer ends up paying £9 and the waiter keeps £2, making £29", but the £2 should be subtracted from what the customers pay, not added.
14. There are two solutions. In the first solution, William and Arthur cross together first, taking 2 minutes, and William then returns with the torch, making 3 minutes in total. Charlie and Benedict cross next, taking 13 minutes in total so far. Arthur takes the torch back (15 minutes) and then finally crosses with William (17 minutes). The second solution is nearly the same, but Arthur makes the first return with the torch.
15. It's impossible, since four odd numbers will always add up to an even number.

16. The cowboy borrows his neighbour's horse, giving him a total of 12. He gives 6 horses to the oldest son, 3 horses to the middle son, and 2 horses to the youngest son. Then he gives the spare horse back to the neighbour.
17. Fill the 3-litre jar and tip all the water into the 5-litre jar. This leaves a 2-litre space in the top of the 5-litre jar. Fill the 3-litre jar again, and pour as much as possible into the 5-litre jar to fill it. There's now 1 litre of water left in the 3-litre jar. Empty the 5-litre jar, then pour the 1 litre of water from the 3-litre jar into it. Fill the 3 litre jar once more and tip the water into the 5-litre jar to make 4 litres. Easy!
18. $64 \times 15625 = 1,000,000$. You can work this out by halving 1,000,000 six times.
19. Yes. This is how you do it:
Open and close first link:

Open and close second link:

Open and close third link to create a circle.
20. Write the letter S to make "SIX".
21. 1113213211. If you read this out loud, it describes the line above in words: "One one, one three, two ones, three twos, one one".
22. 17 ostriches and 13 camels

PAGE 44–45: SQUARE AND TRIANGULAR NUMBERS

Prisoners' puzzle

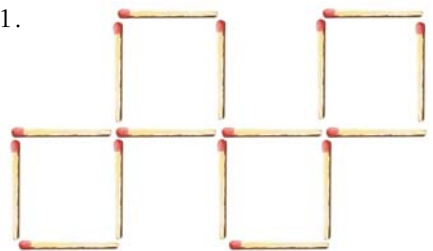
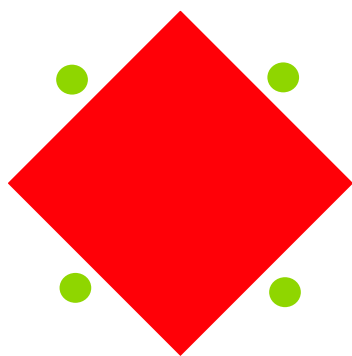
Only prisoners in rooms with a square number on the door escape: 1, 4, 9, 16, 25, 36, and 49.

PAGE 46–47: PASCAL'S TRIANGLE

The road from A to B

There are 56 ways. The numbers form Pascal's triangle on its side.

PAGE 54–55: SHAPES WITH 4 SIDES

1. 
2. 
3. 14. The diagonal line is the same length as the radius of the circle (7+7).
4. 
5. The journey takes as long as it would take to go to C if there was no wind: 30 minutes. This is how pilots actually plan their trips.

